



Trinity College Dublin
Coláiste na Tríonóide, Baile Átha Cliath
The University of Dublin



Courses

Menu

Solar Energy Conversion and Application - Micro-credential

5 ECTS | Level 9

12 weeks

5 Places

Overview

Course Details

Fees

Admissions Requirements

Register your Interest

[Apply](#)

This course qualifies for a 80% subsidy under the HCI Fees Subsidy. Please refer to the fees section for details and eligibility criteria.

Who is this Micro-credential for?

- This micro-credential has been developed to benefit engineers, who are interested in learning more solar energy conversion processes and applications of solar energy systems, architects interested in building integrated solar energy technologies, professionals in local authorities and in NGOs, who are interested in the growing solar industry in Ireland.

Why should I take this Micro-credential?

- This micro-credential will give you an extended foundation of the main concepts of solar energy conversion and applications to enable you to practically apply your new knowledge in the field as well as in research and development.
- Learners will advance their professional competencies through an in-depth focus on how solar cells work up to balance of system components and the design of large solar farms.
- By completing the micro-credential, professional learners will be facilitated to identify and to address these topics and will be enabled to progress their individual career goals in the solar energy industry.
- Our Micro-credentials are all CPD approved by Engineers Ireland

When can I start this Micro-credential?

- There is one intake per year in September

What do I need to be accepted onto this Micro-credential? – Entry requirements & Prerequisite documents

- Applicants must have a 2.1 grade in a level 8 degree in Engineering or a cognate discipline.
- CV & Transcript (2:1)

What will I learn?

- This micro-credential introduces a range of topics in the advanced physics and technology of solar energy conversion and materials, devices, and applications.

Participants will gain an in-depth knowledge of current advances in solar energy, principles of operation of solar thermal and photovoltaic devices, technological challenges, and their applications, as well as an introduction into next generation technologies.

- The aim of this micro-credential is to give the students an extended foundation of the main concepts of solar energy and to enable them to practically apply their knowledge in research and development of solar energy technologies as well as in the design of domestic and solar farm installations.

What will I do?

- In this micro-credential you will attend weekly 2-hour lectures. These will be classroom based but will be live streamed and recorded for those that prefer online delivery. These lectures will advance your professional competencies through an in-depth focus on how solar cells work up to the balance of system and applications.
- One hour each week will be dedicated to group work. In groups of 3-4, you will design solar thermal and photovoltaic systems for specified global locations. Design specifics, scale and selection of software will be decided in class. Optimised solar energy system selection will be determined along with economic analysis and local policy implications. A group presentation will be given, and a report will be produced by you and your group.

How will this Micro-credential be delivered?

This micro-credential will be delivered in a blended format (in-person/online) with all lectures recorded 'live' while also giving the option for in-person lectures.

- Weekly lectures will be live streamed and recorded.
- Weekly groupwork will be dedicated to design a solar

photovoltaic and thermal installation for different geographic locations.

How is this Micro-credential assessed?

- It will be assessed through a group project due in Week 6, worth 50% and a 2-hour examination at the end of term also worth 50%.



Course Coordinator

Daniel Wearen - wearend@tcd.ie

Course Director

Sarah McCormack

Closing Date

12th August

Admission Requirements

- A grade 2.1 level 8 degree in Engineering or a cognate discipline / or satisfy the course coordinator that they can successfully complete the micro-credential programme based on evidence of experience working in the field.
- Up to date Curriculum Vitae

Course Fees

[Click here](#) for a full list of postgraduate fees

HCI Fees Subsidy - Candidates who meet the eligibility criteria may qualify for a 80% fee subsidy, subject to the availability of subsidized places. For eligibility details, please refer to the Eligibility Criteria - <https://www.tcd.ie/courses/microcredentials/hci-eligibility/>

Subsidised Course fee for eligible candidates applying for micro-credential "Solar Energy Conversion and Application" in academic year 2024/25 will be 312 euro.

Apply

To apply, click on the relevant Apply Link below

[Solar Energy Conversion and Application - MC Modul - Module](#)
12/Aug/2025

Get in Touch

Email

General Enquiries: micro-credentials@tcd.ie

Course Coordinator: Daniel Wearen - wearend@tcd.ie

Course Director: Prof Sarah Mc Cormack -
mccorms1@tcd.ie

Register Your Interest

Register your interest at Trinity College Dublin, the University of Dublin.

As Ireland's leading university, we offer a wide range of postgraduate programmes at masters, diploma and certificate level.

Trinity is an international university steeped in history, with a reputation for excellence in education, research and innovation.

[REGISTER YOUR INTEREST](#)

[COURSES](#) [MICRO-CREDENTIALS](#) [BY SCHOOL](#)
[MICRO CREDENTIALS - ENGINEERING](#)
[SOLAR ENERGY CONVERSION AND APPLICATION - MICRO-CREDENTIAL](#)

Updated 8 April 2025

Trinity College Dublin, The University of Dublin.
College Green, Dublin 2, Ireland

T: +353 1 896 1000

Trinity Associations and Charters

[Accessibility](#)

[Cookie policy](#)

[Privacy](#)

[Cookies Settings](#)

[Disclaimer](#)

[Contact](#)

[T-Net](#)